

Analyzing the Impact of Social Media Sentiment on Investor Herding Behavior in Financial Markets Using Agent-Based Modeling

M. D. K. N. Dharmakeerthi[†], T. R. S. K. Fernando[‡], Bhagya Gunarathna[§], D. T. K. S. Adhikari[¶], and A. J. M. M. P. Bandara^{||}

Project advisor: Dr. S. C. Surasinghe[#]

Abstract. This study develops an agent-based model (ABM) to investigate whether social media sentiment drives investor herding and reproduces two well-documented stylized facts of real financial markets: fat-tailed return distributions and volatility clustering. The model, implemented in NetLogo, simulates a heterogeneous population of fundamentalist, optimist-chartist, and pessimist-chartist investors whose trading decisions are shaped by fundamental value, price trends, peer imitation, and a sentiment signal extracted from cryptocurrency tweets using the FinBERT transformer model. Six BehaviorSpace experiments (570 total simulation runs) were used to compare scenarios with and without sentiment, to sweep the herding-strength and vision-radius parameters, and to validate the model against real Bitcoin price and sentiment data. The model successfully reproduced fat tails and volatility clustering across all endogenous-dynamics scenarios. Contrary to the initial hypothesis, however, the strongest fat tails and volatility clustering emerged from the no-sentiment baseline scenario, driven by a peer-imitation (crowd-following) mechanism rather than by sentiment; herding-strength showed only a modest, though statistically significant, positive correlation with volatility ($r = 0.345$, $p = 4.27 \times 10^{-8}$), while vision-radius showed no comparable effect. These results suggest that direct behavioral imitation among investors, rather than sentiment content alone, is the dominant driver of emergent market instability in this model.

1. Introduction. Cryptocurrency markets are widely perceived as highly susceptible to social-media-driven speculation, yet the relative importance of sentiment content versus direct investor imitation in generating market instability remains an open empirical and theoretical question. This project uses an agent-based model to study investor herding behavior in financial markets, using Bitcoin as the empirical anchor. The model simulates a population of investors split into three behavioral types: fundamentalists, who trade based on the gap between price and a perceived fundamental value, and optimist and pessimist chartists, who react to sentiment and to the behavior of other investors around them. Sentiment itself is modeled as something that spreads through the population via tweet agents: individual tweets carrying sentiment, reach, and engagement values are introduced into the simulation, and investors pick up sentiment from tweets within their local vision radius, weighted by each tweet's influence. Price is not read directly from real market data; instead, it emerges endogenously each tick from the imbalance between buyers and sellers, which allows the model to test whether realistic price dynamics – fat tails, volatility clustering, herding spikes – can arise purely from agent interaction rather than being hard-coded.

[†]Student Index s16680, Department of Mathematics, Faculty of Science, University of Colombo, Group 04 .

[‡]Student Index s16688, Department of Mathematics, Faculty of Science, University of Colombo, Group 04 .

[§]Student Index s16694, Department of Mathematics, Faculty of Science, University of Colombo, Group 04 .

[¶]Student Index s16664, Department of Mathematics, Faculty of Science, University of Colombo, Group 04 .

^{||}Student Index s16670, Department of Mathematics, Faculty of Science, University of Colombo, Group 04 .

[#]Department of Mathematics, Faculty of Science, University of Colombo.

To test this, six BehaviorSpace experiments were run: three that compare distinct information environments (no sentiment, a scripted sentiment shock, and pure random sentiment) to isolate what herding actually depends on; two parameter sweeps testing how sensitive market volatility and herding intensity are to herding-strength and vision-radius; and one final run driven entirely by real historical Bitcoin sentiment and price data, to check whether the patterns observed under synthetic conditions also hold against actual market behavior. The remainder of this report is organized as follows. [Section 4](#) describes the model implementation and NetLogo interface. [Section 2](#) describes the data preprocessing pipeline used to construct the sentiment inputs. [Section 3](#) describes the experimental design. [Section 5](#) presents the results, [Section 6](#) discusses and interprets these results in light of the existing literature, [Section 7](#) documents data and code availability, and [Section 8](#) details each member’s individual contribution.

2. Data Preprocessing Methodology. The empirical inputs for the agent-based model were constructed using the Cryptocurrency Tweets dataset. The raw dataset contained tweet text, publication dates, likes, retweets, and follower counts of the tweet authors. The preprocessing pipeline was implemented in Python using the pandas, NumPy, PyTorch, Transformers, and yfinance libraries.

First, the tweet date variable was converted into a consistent datetime format. Observations with invalid dates or missing tweet text were removed. Engagement variables, including likes, retweets, and follower counts, were converted into numeric format. Invalid or missing engagement values were replaced with zero, while negative values were restricted to zero.

Tweet text was standardized by removing URLs, user mentions, punctuation, and unnecessary whitespace. Hashtag symbols were removed while retaining the associated words, and all text was converted to lowercase. Tweets containing fewer than 15 characters after cleaning were removed because they were unlikely to provide sufficient textual information for reliable sentiment classification. Duplicate observations containing the same cleaned text and date were also removed to prevent repeated tweets from having an excessive influence on the sentiment estimates.

Sentiment classification was performed using the ProsusAI/finbert transformer model [1]. For each tweet j , FinBERT produced positive, negative, and neutral class probabilities. A continuous tweet-level sentiment score was calculated as follows:

$$(2.1) \quad s_j = P_j(\text{positive}) - P_j(\text{negative})$$

The score s_j is bounded between -1 and 1 . A value close to 1 represents strongly positive sentiment, while a value close to -1 represents strongly negative sentiment. A value close to zero indicates neutral or mixed sentiment.

Because the FinBERT classification stage required a relatively long processing time, an intermediate checkpoint file was created during sentiment scoring, allowing the processing stage to be resumed if execution was interrupted.

An engagement weight was assigned to each tweet as follows:

$$(2.2) \quad w_j = \text{likes}_j + \text{retweets}_j + 1$$

The addition of one ensured that tweets with no recorded likes or retweets still received a

positive weight. The daily market sentiment score was then calculated as follows:

$$(2.3) \quad S_t = \frac{\sum_{j \in t} s_j w_j}{\sum_{j \in t} w_j}$$

Here, S_t represents the engagement-weighted sentiment score for day t . Historical Bitcoin closing-price data were obtained using the yfinance Python package. The daily Bitcoin price observations were merged with the daily sentiment series using the corresponding calendar date, producing a combined dataset containing the aggregate daily sentiment score and the historical Bitcoin closing price.

A second NetLogo input dataset was created to represent selected tweets as social-media agents. For each available date, the ten tweets with the highest calculated influence were selected, where influence was based on the engagement received by the tweet and the follower count of its author. Reach and engagement variables were normalized to the interval $[0, 1]$ so that variables measured on different original scales could be used consistently within the NetLogo model.

Before the final files were imported into NetLogo, an integration audit was performed, checking the required column names, missing values, sentiment-score ranges, normalized reach and engagement ranges, sequential tick values, and consistency between the tick values in the two final NetLogo input datasets.

2.1. Generated Data Files. The Python preprocessing pipeline generates six CSV files. Four files are intermediate outputs used for transparency, recovery, and validation. The remaining two files are the final inputs imported into the NetLogo model ([Table 1](#)).

Table 1
CSV files generated by the preprocessing pipeline.

File	Type	Description	NetLogo Input
tweets_cleaned.csv	Intermediate	Tweet observations retained after date validation, missing-value treatment, text cleaning, short-text removal, numeric conversion, and duplicate removal.	No
tweets_scored_checkpoint.csv	Intermediate	Partially completed FinBERT sentiment results, allowing processing to resume if notebook execution is interrupted.	No
tweets_scored.csv	Intermediate	Cleaned tweets, FinBERT class probabilities, and continuous tweet-level sentiment scores.	No
daily_sentiment.csv	Intermediate	Engagement-weighted daily sentiment score, tweet count, and simulation tick.	No
daily_sentiment_with_btc.csv	Final	Combines daily aggregate sentiment with historical Bitcoin closing prices; provides the daily market-level inputs required by the NetLogo model.	Yes
tweet_agents.csv	Final	Selected tweet agents with sentiment score, normalized reach, normalized engagement, and corresponding simulation tick.	Yes

2.2. Final NetLogo Input Variables. The final two CSV files are directly imported into the NetLogo model. Their principal variables are summarized in [Table 2](#).

ANALYZING THE IMPACT OF SOCIAL MEDIA SENTIMENT ON INVESTOR HERDING BEHAVIOR IN FINANCIAL MARKETS USING AGENT-BASED MODELING

Table 2

Variables contained in the final NetLogo input files.

Input File	Variable	Description
daily_sentiment_ with_btc.csv	date	Calendar date associated with the daily observation.
	S_t	Engagement-weighted daily sentiment score bounded between -1 and 1 .
	tick	Sequential simulation-day identifier used to align the dataset with NetLogo time.
	tweet_count	Number of cleaned tweets available for the corresponding date.
tweet_agents.csv	btc_close	Historical Bitcoin closing price associated with the corresponding date.
	tick	Simulation tick at which the tweet-agent information is introduced into the model.
	tweet_sentiment	Continuous FinBERT sentiment score bounded between -1 and 1 .
	reach	Normalized measure of the tweet author’s follower count, bounded between 0 and 1 .
	engagement	Normalized measure based on the likes and retweets received by the tweet, bounded between 0 and 1 .

3. Experimental Design and Data Collection. Six experiments were conducted using NetLogo’s BehaviorSpace tool to examine the model’s performance across a variety of sentiment conditions and to test the model’s sensitivity to key parameters. Each parameter combination was replicated 30 times, resulting in 570 individual simulation runs for the entire experimental suite. Replication is important because the model is stochastic: investor types, starting positions, and noise terms are all randomly generated at the beginning of each run, so a single run is not informative on its own. Averaging over 30 runs per condition removes much of this randomness and lets the underlying effect of each scenario or parameter emerge.

In the first three experiments, all other factors were held constant and only the mechanism used to generate sentiment was varied, in order to assess the impact of the information environment in which the market operates. In Experiment 01 (Baseline), no external sentiment data was added to the model, isolating the trading dynamics. Experiment 02 (Sentiment Shock) was designed to observe the population’s response to a sudden, predictable, and large change in sentiment: sentiment stayed at zero until tick 50, then jumped to $+0.8$ for 50 ticks, then dropped to -0.6 for the rest of the run. In Experiment 05 (Random Sentiment), the simulated shock was simply random noise, so that if herding still emerged, it would be attributable to agents reacting to one another rather than to sentiment acting as a reliable signal. All three scenarios ran for 150 ticks with herding-strength fixed at 0.6 and vision-radius fixed at 8 .

Experiments 03 and 04 were parameter sweeps rather than scenario comparisons, and both were conducted on top of the sentiment-shock condition rather than the baseline condition, since that is where a herding effect is most likely to be detectable; against a flat, no-signal baseline, a parameter sweep would reveal little about that parameter’s effect on herding. In

Experiment 03 (Herding Sensitivity), eight values of herding-strength were tested (0.2 to 0.9 in steps of 0.1) with vision-radius fixed at 8, for a total of 240 runs; the purpose was to determine whether the herding index could be increased further by strengthening herd-following behavior, or whether it would reach a plateau. Experiment 04 (Vision Radius Sweep) followed the same design with herding-strength fixed at 0.6 and seven values of vision-radius tested, for a total of 210 runs, asking whether letting agents see further (i.e., observe more tweets at once) affects the rate at which sentiment spreads through the population or the strength of the resulting herding effect.

The final experiment (06: Dataset) replaced all synthetic sentiment generators with the real sentiment and price data of Bitcoin gathered for this project (Section 2). Because of the limited length of the dataset (36 trading days), herding-strength and vision-radius were kept at their default values (0.6 and 8, respectively) and the run was limited to 36 ticks. The primary goal of this experiment was to test whether the patterns that emerge under synthetic conditions – fat tails, volatility clustering – also emerge when the model is driven by real market sentiment rather than an artificial signal.

The same six reporters (price, buyers, sellers, herding intensity, volatility, and sentiment) were logged at every tick in all six experiments, producing a complete time series of reporters rather than only an end-of-run snapshot. The analysis in Section 5 is based on these tick-level time series.

4. Model Implementation. The proposed agent-based model was implemented in NetLogo to investigate how social media sentiment influences investor herding behavior in the cryptocurrency market. The model consists of investor agents and tweet agents. Investor agents are categorized into fundamentalists, optimistic chartists, and pessimistic chartists. During each trading tick the model updates external sentiment, generates tweet agents, computes local sentiment, executes investor trading decisions, and updates the market price according to aggregate demand. The simulation continuously records price, buyers, sellers, herding intensity, volatility, returns, and sentiment, providing the basis for the experimental analysis presented in Section 5.

4.1. Investor Demand and Price Formation. Fundamentalist agents’ demand depends solely on the gap between the current price and the model’s fundamental price, with no sentiment or crowd-following component; fundamentalists never participate in herding behavior. Optimist and pessimist chartists’ demand, by contrast, includes two additive components: a sentiment-driven term and a crowd-following term,

$$(4.1) \quad D_{i,t} = \underbrace{\gamma_i s_{i,t}^{\text{local}}}_{\text{sentiment-driven}} + \underbrace{h \cdot C_t}_{\text{crowd-following}},$$

where $s_{i,t}^{\text{local}}$ is the local sentiment perceived by agent i at tick t (aggregated from nearby tweet agents within the agent’s vision radius, weighted by each tweet’s influence), γ_i is the agent’s sensitivity to sentiment, h is the herding-strength parameter, and C_t is the crowd-bias at tick t : the net buy/sell imbalance among all investors within the same tick, independent of any external information. Because the crowd-following term in (4.1) is active regardless of whether an external sentiment signal is present, it constitutes a self-referential feedback loop

ANALYZING THE IMPACT OF SOCIAL MEDIA SENTIMENT ON INVESTOR HERDING BEHAVIOR IN FINANCIAL MARKETS USING AGENT-BASED MODELING

in which agents react to other agents' decisions rather than solely to exogenous information. The aggregate excess demand across all investors determines the change in simulated market price each tick, subject to a lower price floor ($\text{price} \geq 1$) that prevents negative or nonsensical prices.

The herding index at tick t is computed as $|B_t - S_t|/N$, where B_t and S_t are the numbers of buying and selling investors, respectively, and N is the total investor population (100 investors in this study). Market volatility and returns are computed from the tick-level price series.

4.2. NetLogo Interface. Figures 1 to 3 show the principal interface components used to configure and monitor the simulation, while Figures 4 to 7 show representative real-time plots used to validate model behavior during execution.

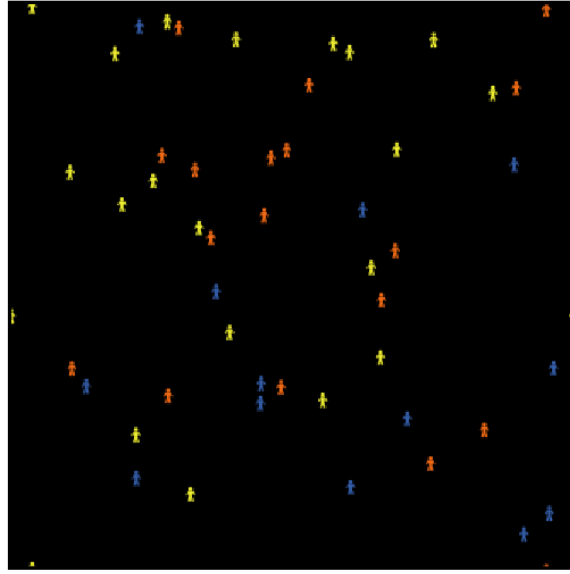


Figure 1. NetLogo world view showing the spatial arrangement of investor and tweet agents.

The graphical behavior of the model was validated through repeated simulation runs under different parameter settings. Investor colors were observed to confirm that agent states changed dynamically according to market decisions and behavioral categories, and the world view, price plot, buyers-versus-sellers plot, sentiment plot, herding-intensity plot, volatility monitor, and return monitor were checked for consistency with the internal model state. This visual verification helped confirm that agent interactions, sentiment transmission, and price dynamics were operating as intended, including emergent patterns such as temporary buying waves, selling pressure, high-herding periods, and price fluctuations caused by supply-and-demand imbalances.

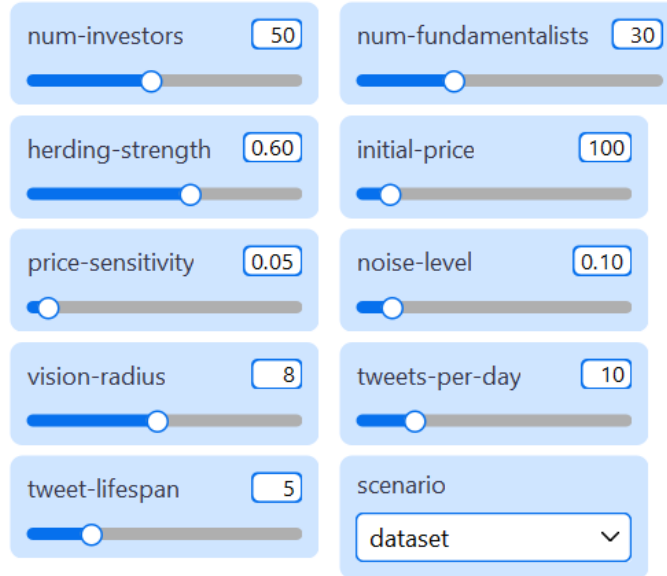


Figure 2. *Simulation parameter controls, including herding-strength, vision-radius, and scenario-selection sliders/switches.*

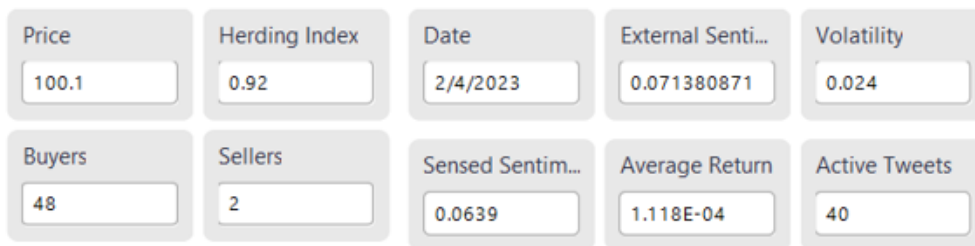


Figure 3. *Real-time performance monitors used to validate model execution.*



Figure 4. Real-time price-over-time plot.

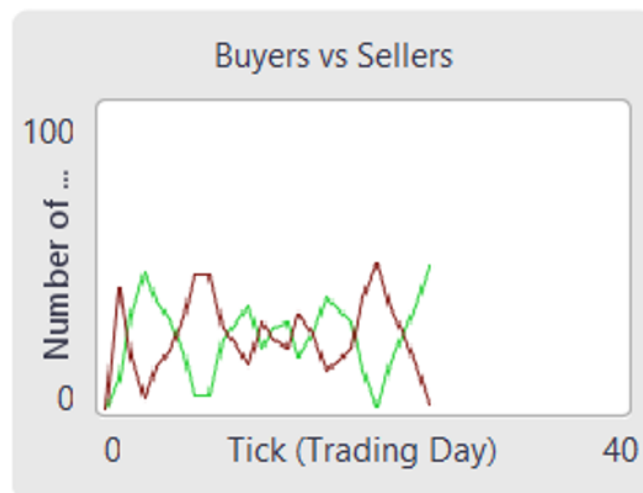


Figure 5. Real-time buyers-versus-sellers plot.

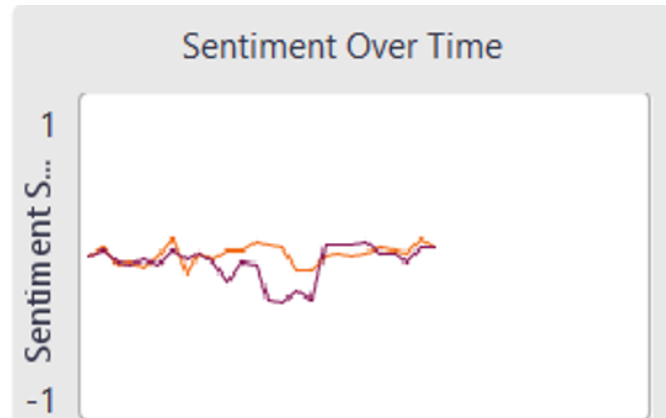


Figure 6. *Real-time sentiment-over-time plot.*

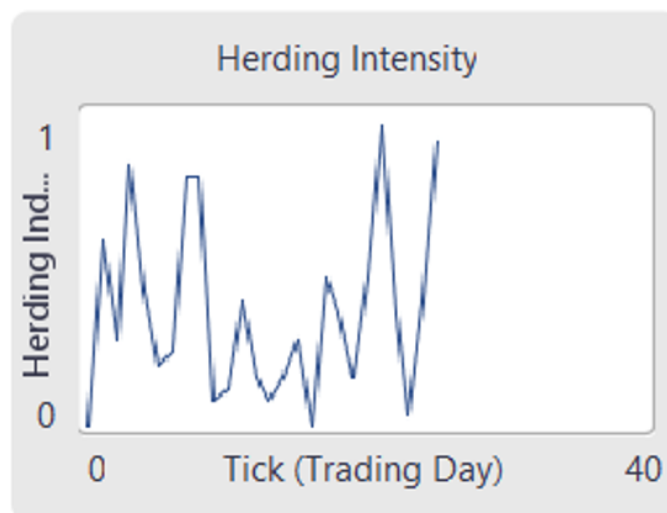


Figure 7. *Real-time herding-intensity plot.*

5. Results. The agent-based herding model was evaluated across six simulation scenarios (Section 3), each producing tick-level records of price, return, volatility, herding index, and buyer/seller counts. The 01-Baseline scenario (150 ticks, herding-strength = 0.6, vision-radius = 8, no external sentiment, 30 replicate runs) establishes reference market dynamics under default parameters. SentimentShock introduces an exogenous, scripted sentiment perturbation for direct comparison against the baseline (same fixed parameters, 30 runs). HerdingSensitivity sweeps the herding-strength parameter across eight values (0.2–0.9) on top of the sentiment-shock condition, with 30 replicate runs per value (240 runs total), to test its effect on volatility, while VisionRadiusSweep performs an analogous sweep over seven vision-radius values (210 runs total). RandomSentimentComparison replaces the model’s structured sentiment mechanism with random noise as a control (30 runs), and Dataset contains the 36-tick run driven by the empirical Bitcoin sentiment and price series. Results are organized below by research question, beginning with time-series comparisons, followed by cross-scenario volatility comparisons, parameter-sensitivity analyses, and finally the distributional and temporal properties of returns.

5.1. Price Dynamics and Herding Behavior: Baseline vs. Sentiment Shock. Average price trajectories for the Baseline and SentimentShock scenarios diverge visibly over the simulated trading period (Figure 8). The Baseline series fluctuates around its starting level with intermittent sharp movements, whereas the SentimentShock series shows a comparable overall pattern with a numerically lower average price (94.589 vs. 117.256; Table 3) and numerically higher mean volatility (0.026 vs. 0.022; Table 3). Neither difference reached statistical significance at the run level (Welch’s $t = 1.25$, $p = .221$ for price; $t = -1.08$, $p = .285$ for volatility, $n = 30$ runs per scenario), indicating that these two scenarios are not reliably distinguishable on price or volatility level alone despite the visible divergence in Figure 8.

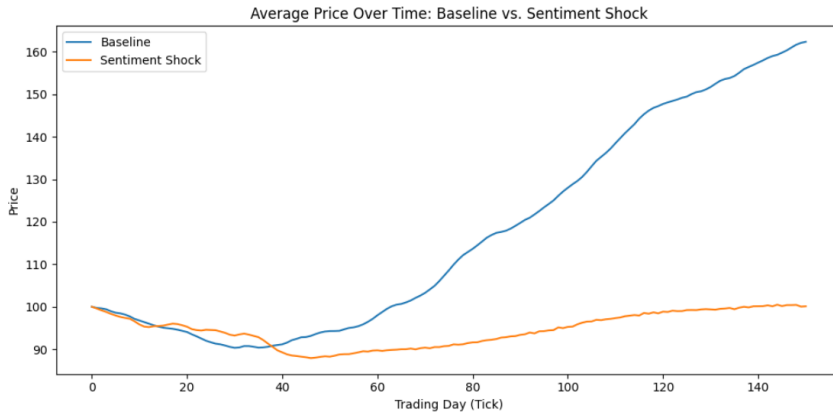


Figure 8. Average price over time, Baseline vs. Sentiment Shock.

The herding index follows a broadly similar time course in both scenarios (Figure 9), with mean values of 0.724 for Baseline and 0.664 for SentimentShock (Table 3), indicating that the sentiment perturbation is associated with a modest reduction in average herding intensity relative to baseline conditions.

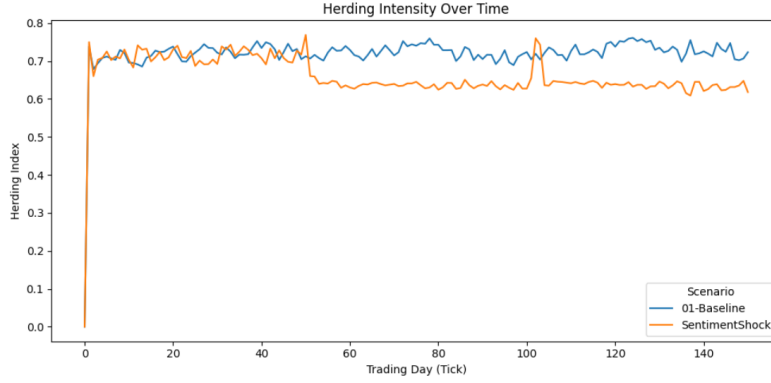


Figure 9. Herding index over time, Baseline vs. Sentiment Shock.

5.2. Market Volatility Across Scenarios. Volatility distributions differ substantially across the four scenarios compared directly: Baseline, SentimentShock, RandomSentimentComparison, and Dataset (Figure 10). The two simulated scenarios with endogenous sentiment/herding dynamics (Baseline and SentimentShock) display a wider spread of volatility values, including a greater number of extreme values, than RandomSentimentComparison and Dataset. A one-way ANOVA confirms that mean volatility differs significantly across the four scenarios ($F = 7.64$, $p = .0001$), consistent with the visibly tighter volatility distributions of RandomSentimentComparison and Dataset relative to Baseline and SentimentShock. Mean volatility is lowest for RandomSentimentComparison (0.013) and highest for SentimentShock (0.026), with Baseline (0.022) and Dataset (0.016) falling between these values (Table 3).

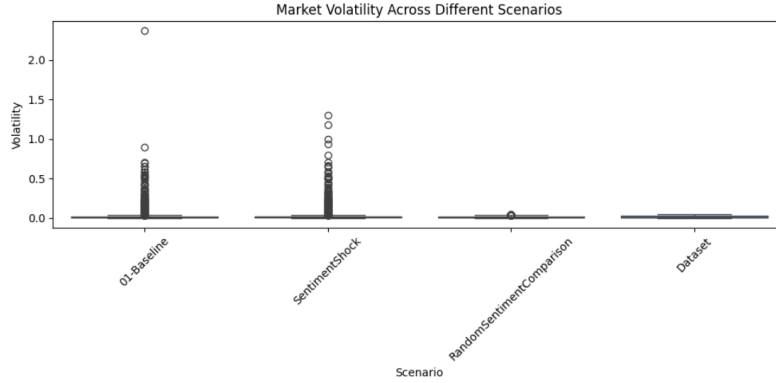


Figure 10. Market volatility distributions across scenarios.

ANALYZING THE IMPACT OF SOCIAL MEDIA SENTIMENT ON INVESTOR HERDING BEHAVIOR IN FINANCIAL MARKETS USING AGENT-BASED MODELING

Table 3

Descriptive statistics across simulation scenarios (per-run means, n = number of runs; 95% CI computed on per-run averages).

Scenario	Mean Price [95% CI]	Mean Volatility [95% CI]	Mean Herding Index [95% CI]	n
01-Baseline	117.256 [82.28, 152.24]	0.022 [0.017, 0.026]	0.724 [0.712, 0.736]	30
SentimentShock	94.589 [81.88, 107.30]	0.026 [0.019, 0.034]	0.664 [0.660, 0.669]	30
HerdingSensitivity	102.272 [98.20, 106.35]	0.022 [0.019, 0.024]	0.662 [0.660, 0.664]	240
VisionRadiusSweep	97.571 [92.95, 102.19]	0.025 [0.022, 0.028]	0.668 [0.666, 0.670]	210
RandomSentimentComparison	100.167 [99.77, 100.57]	0.013 [0.012, 0.013]	0.640 [0.633, 0.646]	30
Dataset	99.977 [99.73, 100.23]	0.016 [0.015, 0.017]	0.678 [0.672, 0.685]	30

5.3. Effect of Herding Sensitivity on Volatility. Within the HerdingSensitivity experiment, a correlation heatmap of herding-strength against price, volatility, herding-index, and buyer/seller counts (Figure 11) shows herding-strength most strongly associated with volatility relative to the other variables in the matrix. Using per-run averaged values to avoid pseudo-replication across ticks within a run, herding-strength and mean volatility show a positive Pearson correlation of $r = 0.345$ ($p = 4.27 \times 10^{-8}$), indicating a statistically significant, moderate positive association.

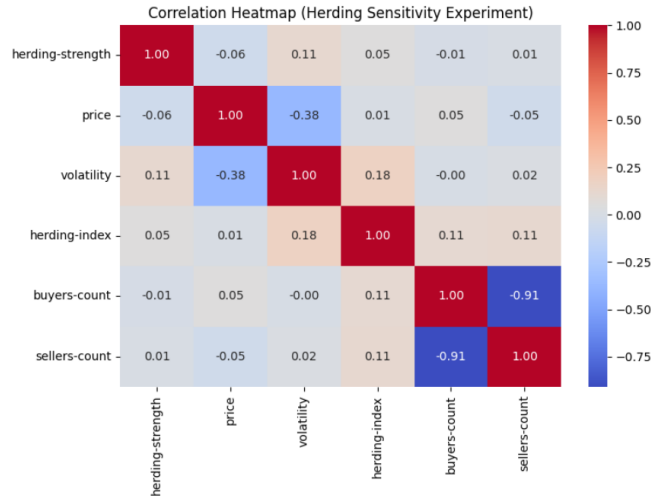


Figure 11. Correlation heatmap of key variables, Herding Sensitivity experiment.

The boxplot distribution of volatility across sampled herding-strength levels (Figure 12) shows volatility spread and median values shifting across the sweep, consistent with the positive correlation reported above.

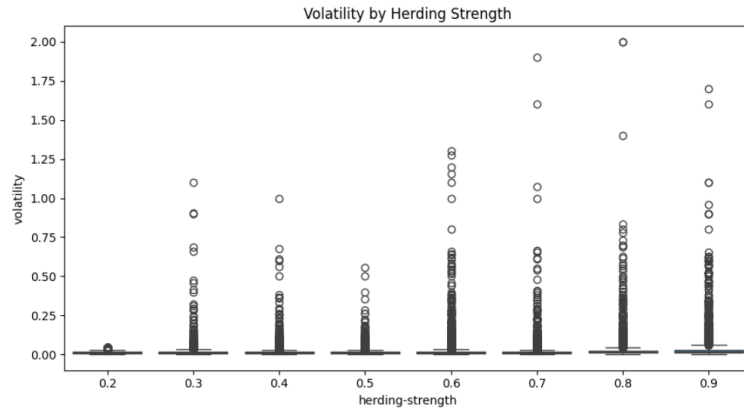


Figure 12. *Volatility by herding-strength level, Herding Sensitivity experiment.*

5.4. Effect of Vision Radius on Volatility. In the VisionRadiusSweep experiment, volatility is plotted against the range of tested vision-radius values (Figure 13). Volatility varies across vision-radius levels, with the mean volatility for this scenario (0.025; Table 3) close to that observed in HerdingSensitivity (0.022) and Baseline (0.022), and above that of the RandomSentimentComparison control (0.013).

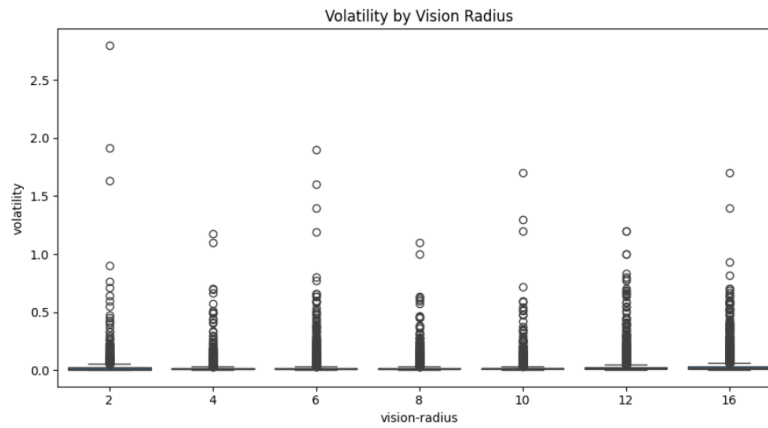


Figure 13. *Volatility by vision-radius level, Vision Radius Sweep experiment.*

5.5. Distributional Properties of Returns. Excess kurtosis of per-tick returns was computed for each scenario as an indicator of fat-tailed return distributions (Table 4, “All ticks” column). Raw excess kurtosis values are markedly elevated in the simulated scenarios with endogenous dynamics (451.827 for Baseline; 300.776 for HerdingSensitivity; 301.921 for VisionRadiusSweep; 122.691 for SentimentShock) relative to RandomSentimentComparison (−0.403) and Dataset (−0.695). After excluding ticks affected by price-floor rebound artifacts (price ≤ 15), excess kurtosis in the simulated scenarios falls substantially but remains positive and above the two control/empirical series (Table 4, “Filtered” column): 24.531 for Baseline, 4.754 for HerdingSensitivity, 4.281 for VisionRadiusSweep, and 2.953 for SentimentShock, compared with −0.695 for Dataset and −0.403 for RandomSentimentComparison.

An outlier check on per-tick returns shows the widest range of extreme values in Baseline, HerdingSensitivity, and VisionRadiusSweep (minimum returns near −0.70 and maximum returns exceeding 2.0), compared to a much narrower range in RandomSentimentComparison and Dataset (returns bounded within approximately ± 0.05). In the Baseline scenario specifically, 14 tick-level observations exceed an absolute return of 0.5, concentrated in a small number of simulation runs.

Table 4

Excess kurtosis of returns by scenario (all ticks vs. filtered to exclude price-floor artifacts).

Scenario	Excess Kurtosis (All ticks)	Excess Kurtosis (Filtered, price > 15)
01-Baseline	451.827	24.531
SentimentShock	122.691	2.953
HerdingSensitivity	300.776	4.754
VisionRadiusSweep	301.921	4.281
RandomSentimentComparison	−0.403	−0.403
Dataset	−0.695	−0.695

5.6. Volatility Clustering. Average per-run autocorrelation of volatility was calculated at lags of 1 and 5 ticks for each scenario (Table 5). The four simulated scenarios with endogenous herding dynamics show substantial positive autocorrelation at both lags, ranging from 0.577 to 0.826 at lag 1 and 0.427 to 0.624 at lag 5. In contrast, RandomSentimentComparison and Dataset show autocorrelation values near zero or slightly negative at both lags (−0.192 to −0.326 at lag 1; −0.029 to 0.010 at lag 5).

Table 5

Average per-run autocorrelation of volatility by scenario.

Scenario	Lag 1	Lag 5
01-Baseline	0.826	0.624
SentimentShock	0.650	0.451
HerdingSensitivity	0.577	0.438
VisionRadiusSweep	0.616	0.427
RandomSentimentComparison	−0.192	0.010
Dataset	−0.326	−0.029

5.7. Summary of Comparative Findings. Across all analyses, the simulated scenarios driven by endogenous herding and sentiment dynamics (Baseline, SentimentShock, HerdingSensitivity, VisionRadiusSweep) consistently show higher mean volatility, higher excess kurtosis of returns, and stronger volatility autocorrelation than the two control/reference series (RandomSentimentComparison and Dataset). Within the parameter-sweep experiments, herding-strength shows a statistically significant positive correlation with mean volatility ($r = 0.345$, $p = 4.27 \times 10^{-8}$), and vision-radius is associated with comparable mean volatility levels to the herding-strength sweep and the Baseline scenario.

6. Discussion and Conclusion.

6.1. Summary of Main Findings. This study aimed to find out whether social media sentiment influences investor herding and whether it can reproduce two common features of real financial markets: fat-tailed returns and volatility clustering. The results partially support this idea, but they also reveal another important finding. As expected, the SentimentShock scenario showed clearer fat tails (excess kurtosis of 2.953, filtered; Table 4) and stronger volatility clustering (lag-1 autocorrelation of 0.650; Table 5) than the RandomSentimentComparison and Dataset scenarios, which did not include a strong sentiment signal or endogenous herding. In the Herding Sensitivity analysis, increasing herding strength had a small but statistically significant positive relationship with average volatility ($r = 0.345$, $p = 4.27 \times 10^{-8}$; Figure 12). This suggests that investors who are more likely to follow others can increase market volatility. On the other hand, changing the vision radius did not have much effect, as the average volatility stayed almost the same across all tested values (Table 3; Figure 13).

An unexpected result was that the Baseline scenario, where external sentiment was always set to zero, produced the strongest fat tails (excess kurtosis of 24.531, filtered; Table 4) and the highest volatility clustering (lag-1 autocorrelation of 0.826; Table 5). This happened because of the model’s crowd-bias mechanism, where investors copied the behavior of others even without any sentiment input. Another interesting finding was that the Dataset scenario, which used real cryptocurrency Twitter sentiment, produced results that were very similar to the RandomSentimentComparison scenario. The excess kurtosis was -0.695 compared with -0.403 (Table 4), suggesting that the daily changes in the real sentiment data were too small to have a noticeable effect beyond random variation. Overall, the findings suggest that while strong or prolonged sentiment can contribute to herding and increased volatility, peer imitation has a much greater influence in this model and is the main reason for the fat-tailed returns and volatility clustering observed.

6.2. Interpretation and Implications.

6.2.1. Mechanisms Underlying Model Behavior. The results can be explained by the way investor demand is calculated in the model (Equation (4.1)). The demand of optimistic and pessimistic chartists is based on two parts: a sentiment term ($\gamma_i \times$ local sentiment) and a crowd-following term (herding strength $\times$ crowd bias). The crowd bias measures the difference between buyers and sellers during the same trading tick, regardless of any external sentiment. Since the crowd-following term is active in every scenario, including the Baseline scenario where sentiment is always zero, investors continue to react to each other’s behavior instead of

outside information. This creates a feedback loop where small differences between buyers and sellers grow larger over time: even a small imbalance caused by chance can quickly become much larger because investors keep copying each other. This is a plausible explanation for why the Baseline scenario produced the strongest fat tails and volatility clustering out of all the scenarios tested (Tables 4 and 5), although, as discussed in Subsection 6.4, the herding measures used here also include fundamentalists, whose synchronized reaction to a shared fundamental-price signal cannot currently be distinguished from genuine peer imitation.

This also explains the results of the parameter analysis. Herding strength controls how much importance investors give to the crowd-following term. As expected, increasing herding strength led to a small but significant increase in average volatility. However, the effect was not very large because the crowd-following behavior was already active even at lower herding-strength values; increasing its value only strengthened a mechanism that was already influencing the market. In contrast, vision radius only changes how far investors can observe sentiment and the positions of other investors. It does not directly increase the weight of the crowd-following behavior, which is likely why it showed little relationship with volatility.

These findings are similar to the “on-off intermittency” behavior described by Lux and Marchesi [5]. They showed that periods of high market volatility can develop from the interactions between investors without any external news or shock. A similar pattern appears in this model. Most of the time the market remains fairly stable, but when enough investors begin following the crowd, volatility increases rapidly. Eventually, fundamentalist traders pull prices back towards their fundamental value, making the market stable again. The price-floor behavior observed during model validation also fits this explanation: strong selling caused prices to fall close to the lower price limit, and the recovery afterwards created unusually large percentage returns, resulting in short periods of extreme volatility.

The similarity between the Dataset and RandomSentimentComparison scenarios can also be explained in this way. The sentiment values taken from the cryptocurrency tweet dataset were much smaller than the extreme values used in the SentimentShock scenario. Since the crowd-following behavior already has a strong influence on investor decisions, these smaller sentiment changes were not strong enough to noticeably affect market behavior. As a result, the real sentiment data produced results that were very similar to random sentiment. This suggests that for sentiment to have a clear impact on herding, it needs to be both strong and persistent enough to compete with the effects of peer imitation.

6.2.2. Connections to Real-World Phenomena and Theoretical Expectations. These findings relate to an important question in behavioral finance: is market instability mainly caused by new information, or by investors copying each other’s behavior? The results of this study suggest that peer imitation has a stronger influence than sentiment itself. This is similar to what has been seen in real financial markets, especially during rapid price increases and crashes in retail-driven stocks and cryptocurrency markets. During these periods, investors often make decisions based on what other traders are doing, such as following price trends or buying because others are buying, rather than because of new information.

The results therefore suggest that sentiment alone may not fully explain herding behavior. Instead, direct imitation between investors appears to play a major role in creating fat-tailed returns and volatility clustering. This supports the idea that market instability can develop

from the interactions between investors themselves, even when there is little or no new external information. This is consistent with the stylized facts of financial markets discussed by Cont [3].

6.3. Comparison to Previous Work. The overall structure of this model is similar to many existing agent-based financial market models. Like the model developed by Hessary and Hadzikadic [4], it includes three types of investors – fundamentalists, optimists, and pessimists – whose trading decisions determine market prices. Their study successfully reproduced two important stylized facts of financial markets, fat-tailed returns and volatility clustering, and found that herding and market volatility were closely related. The results of the present study are generally consistent with these findings. Herding strength had a significant positive relationship with average volatility ($r = 0.345$, $p < .001$), suggesting that stronger herding increases market fluctuations. However, unlike their study, the strongest fat tails in this model appeared in the Baseline scenario, where there was no external sentiment. This suggests that investor interactions alone can generate realistic market behavior, even without sentiment.

This result is also consistent with Lux and Marchesi [5], who showed that fat tails and volatility clustering can emerge naturally from investors copying each other’s behavior, without any external news or information. A similar pattern was observed in this study. The crowd-following mechanism created periods of calm trading followed by sudden bursts of high volatility, before prices gradually returned towards their fundamental value. This suggests that peer imitation is a major driver of market instability.

Finally, the finding that the Dataset scenario produced results similar to random sentiment is consistent with Bokhari [2]. Bokhari argued that social media sentiment has a stronger influence during certain market conditions, particularly bearish markets, rather than at all times. Since the cryptocurrency sentiment data used in this study covered only a short period and did not include a prolonged bearish market, the sentiment signal may not have been strong enough to produce a noticeable effect. This may explain why the real sentiment data behaved similarly to random noise in the simulations.

6.4. Limitations. This study has several limitations that should be considered when interpreting the results.

Price-floor artifact. The model prevents prices from falling below 1. In some cases, especially when strong crowd-following occurred, prices rebounded sharply after reaching this lower limit, creating unrealistically large returns. These extreme values were removed from the analysis, but they show that part of the extreme behavior was caused by the model’s price rule rather than investor behavior.

Difficulty separating herding effects. The model combines social media sentiment and crowd-following in the same demand equation (Equation (4.1)), making it difficult to measure the individual effect of each factor. A model that allows these mechanisms to be switched on or off independently would provide a clearer comparison.

Limited role of fundamentalists. Fundamentalist investors never follow the crowd or respond to sentiment. Because they always trade based on fundamental value, they reduce the overall level of herding in the market, which may have limited the variation in the results.

Weak real-world sentiment data. The sentiment values taken from cryptocurrency tweets were much smaller than those used in the SentimentShock scenario. As a result, it is difficult to

know whether the real sentiment data truly contained little useful information or whether the sentiment signal was simply too weak to compete with the model's crowd-following behavior.

True vs. spurious herding.. The herding measures included both chartists and fundamentalists. Since all fundamentalists react to the same fundamental price, they can make similar trading decisions without actually copying one another. This means that some of the herding measured in the model may reflect investors responding to the same information rather than genuine imitation. Separating these two types of herding would provide a clearer understanding of the model's behaviour.

Limited calibration.. The model was not fully calibrated using real Bitcoin market data. Instead, it was validated by checking whether it reproduced well-known market characteristics such as fat-tailed returns and volatility clustering. Therefore, the model should be viewed as demonstrating a possible explanation for market behavior rather than as a tool for accurate price prediction.

Order of agent decisions.. In NetLogo, agents make decisions in a random order each tick. This means some investors react to earlier decisions made in the same tick, while others do not, which may slightly affect the simulation results.

Population and interaction scale.. The model includes only 100 investors, which is much smaller than similar studies. The smaller population and simplified interaction rules may reduce how closely the simulation matches the behavior of real financial markets.

6.5. Future Directions. Several improvements could be made in future research to strengthen this model and its findings:

- Separate the effects of social media sentiment and crowd-following in the demand equation so that the influence of each mechanism can be measured more clearly.
- Complete the planned Black-it and pyNetLogo calibration using real Bitcoin market data to improve the model's accuracy and make it more representative of real markets.
- Use a longer and more volatile social media sentiment dataset to better determine whether real sentiment has a meaningful effect on market behavior.
- Add a retweet or information-sharing mechanism, allowing investors to spread sentiment through their network, to better represent how information spreads on social media.
- Replace the hard price floor with a smoother price boundary to avoid the unrealistic price rebounds observed in the simulations.
- Increase the number of investors and introduce a more realistic network structure for interactions, similar to previous agent-based financial market studies, to examine whether the findings remain consistent in a larger and more complex market.
- Apply Granger causality analysis to investigate whether herding causes changes in volatility, or whether the relationship works in both directions.
- Separate true and spurious herding by calculating the herding-index and crowd-bias for chartists only, excluding fundamentalists. This would help determine whether the herding observed in the model is mainly caused by peer imitation or by investors independently responding to the shared fundamental price.

6.6. Conclusion. This study investigated how social media sentiment influences investor herding in cryptocurrency markets using an agent-based model. The model included three types of investors – fundamentalists, optimists, and pessimists – who made trading decisions based on both social media sentiment and the behavior of other investors. The model was able to reproduce two well-known characteristics of real financial markets: fat-tailed returns and volatility clustering. This shows that relatively simple trading rules and interactions between different types of investors can produce realistic market behavior.

However, the main finding was different from what was originally expected. The results showed that peer-driven crowd-following had a much greater impact on market behavior than social media sentiment. Even in the Baseline scenario, where there was no external sentiment, the model produced the strongest fat tails and volatility clustering. Increasing herding strength led to a small but significant increase in volatility, while changing the vision radius had little effect. Although extreme sentiment shocks increased market volatility, the real cryptocurrency sentiment data produced results that were very similar to random sentiment, suggesting that the sentiment signal was not strong enough to noticeably influence investor behavior in this model.

These findings suggest that social media sentiment is not the only factor behind investor herding. Instead, the tendency of investors to copy each other’s behavior appears to play a much more important role in creating market instability. This means that monitoring social media sentiment alone may not be enough to understand or predict periods of high market volatility. Future research should continue to separate the effects of sentiment and peer imitation and test the model using larger datasets and more realistic market conditions to better understand how these factors interact in real financial markets.

7. Data and Code Availability. The complete codebase and supporting materials used in this project are available in the project GitHub repository:

https://github.com/nilupuldharmakeerthi-rgb/FM-4054_ABM_Group-04.git

The repository contains the following materials:

- the NetLogo 7.0.3 agent-based model;
- the Python notebook used for data preprocessing and FinBERT sentiment analysis;
- the processed datasets used by the NetLogo model;
- the BehaviorSpace experiment configurations and outputs;
- the Python files used for result analysis and visualization;
- the figures used in the final report; and
- the final project report.

The original Cryptocurrency Tweets dataset is not included in the GitHub repository because of its large file size. The dataset can be obtained from Kaggle:

https://www.kaggle.com/datasets/infscpts/cryptocurrency-tweets?select=Merged_Twitter_Data.IDsRemoved.csv

After downloading the original dataset, users can reproduce the processed datasets by running all cells of the preprocessing notebook in sequential order. Historical Bitcoin closing-price data are downloaded programmatically through the yfinance Python package. The final processed datasets required by the NetLogo model are:

- `daily_sentiment_with_btc.csv`
- `tweet_agents.csv`

These files allow the NetLogo simulations to be reproduced without repeating the computationally intensive FinBERT sentiment-classification stage.

8. Individual Contribution Statement.

M. D. K. N. Dharmakeerthi – s16680. Submodel Implemented or Analyzed. Responsible for developing and validating the external social-media data component of the model. This included preparing the Cryptocurrency Tweets dataset, constructing the daily sentiment and tweet-agent inputs, and integrating the generated CSV files with the NetLogo model. Also checked the compatibility of the dataset column names, variable order, simulation ticks, and numerical ranges with the NetLogo data-import procedures.

Analysis or Visualization Contribution. Developed and executed the Python data-preprocessing pipeline using pandas, NumPy, PyTorch, the Hugging Face Transformers library, and yfinance. The pipeline cleaned and deduplicated the tweet observations, generated FinBERT sentiment scores, calculated engagement-weighted daily sentiment, merged the sentiment series with historical Bitcoin prices, and selected the ten most influential tweets for each date for use as NetLogo tweet agents. Also implemented validation checks for missing values, required column structures, sentiment-score ranges, normalized reach and engagement variables, sequential tick values, and consistency between the final NetLogo input datasets.

Writing Contribution. Wrote the Data Preprocessing Methodology (Section 2) and Data and Code Availability (Section 7) sections of the final report. Also documented the generated CSV files, prepared the repository structure, and contributed to the reproduction instructions included in the GitHub README file.

Understanding of Project. The project investigates how social-media sentiment and

peer influence can generate investor herding and market instability within a simulated cryptocurrency market. Historical tweet text is converted into quantitative sentiment information using FinBERT. The model uses both a daily aggregate sentiment signal and individual tweet agents with different levels of reach and engagement. Fundamentalist investors respond primarily to deviations between the simulated market price and its fundamental value. Chartist investors respond to market trends, social-media sentiment, and the behavior of other investors. The combined buying and selling decisions of the investor agents create excess demand, which subsequently changes the simulated market price. By comparing different experimental scenarios, the model evaluates whether social-media sentiment produces stronger herding, volatility, and price movements than those generated under alternative market conditions.

T. R. S. K. Fernando – s16688. Submodel Implemented or Analyzed. The main technical contribution focused on the implementation and validation of the core investor-behavior submodel within the NetLogo simulation (Section 4). This included the development of heterogeneous investor agents classified as fundamentalists, optimistic chartists, and pessimistic chartists. Each investor type was assigned a distinct decision-making rule so that the simulated market could represent different behavioral responses to price movements, perceived sentiment, and herding pressure. The interaction between investor agents and tweet agents was also incorporated to ensure that social media information could influence local trading decisions.

Analysis or Visualization Contribution. The simulation workflow was tested from the setup stage to the completion of each trading tick, including checking the loading of the sentiment and tweet datasets, the generation and aging of tweet agents, the calculation of local sentiment, the execution of investor decisions, and the updating of market indicators. The graphical behavior of the model was validated through repeated simulation runs under different parameter settings; the world view, price plot, buyers-versus-sellers plot, sentiment plot, herding-intensity plot, volatility monitor, and return monitor were checked for consistency with the internal model state. Produced the NetLogo interface screenshots used in Subsection 4.2.

Writing Contribution. Prepared the Model Implementation introduction and NetLogo interface documentation (Section 4) for the report, explaining the role of investor agents, tweet agents, sentiment integration, and price formation.

Understanding of Project. The overall project investigates how social media sentiment can influence investor herding behavior and generate market-level effects in a cryptocurrency trading environment. FinBERT-derived sentiment scores provide the external information signal, while tweet agents distribute this information locally across the simulation space. Investors do not all react in the same way; instead, their responses depend on their assigned behavioral type, perceived local sentiment, market conditions, and herding pressure. The combined actions of these heterogeneous investors produce aggregate buying and selling pressure, which then determines simulated price changes. The model therefore provides a controlled environment for examining how social media information, behavioral diversity, and local interactions can contribute to collective market behavior.

Bhagya Gunarathna – s16694. Submodel Implemented or Analyzed. Designed and executed the BehaviorSpace experimental suite for the investor-herding ABM (Section 3). Configured six automated experiments: two baseline scenario comparisons (no-sentiment baseline and pure random-sentiment control), one scripted sentiment-shock condition, two parameter sensitivity sweeps (herding-strength across 8 values, vision-radius across 7 values), and one validation run against real Bitcoin sentiment and price data, totaling 570 individual simulation runs across 30 repetitions per condition.

Analysis or Visualization Contribution. Exported and structured all raw BehaviorSpace table outputs (six CSV files, one per experiment) and verified run counts, parameter values, and tick ranges against the experimental design before handing off to the data analysis stage. Ran an initial Python summary script in Google Colab to compute preliminary distributional statistics (kurtosis, lag-1 autocorrelation) across scenarios, producing a consolidated `results_summary.csv`.

Writing Contribution. Wrote the Experimental Design and Data Collection section (Section 3) of the report, detailing the design, parameters, and rationale of all six experiments.

Understanding of Project. This project uses an agent-based model to study investor herding behavior in financial markets, using Bitcoin as the empirical anchor. The model simulates a population of investors split into three behavioral types: fundamentalists, who trade based on the gap between price and a perceived “true” value, and optimists/pessimists, who react to sentiment and to what other investors around them are doing. Sentiment itself is modeled as something that spreads through the population via tweet agents: individual tweets carrying sentiment, reach, and engagement values are spawned into the simulation space, and investors only pick up sentiment from tweets within their local vision radius, weighted by that tweet’s influence. Price is not read directly from real market data; instead, it emerges endogenously each tick from the imbalance between buyers and sellers, which lets us test whether realistic price dynamics like fat tails, volatility clustering, and herding spikes can arise purely from agent interaction rather than being hard-coded. To test this, six experiments were run: three that compare distinct information environments (no sentiment, a scripted sentiment shock, and pure random sentiment) to isolate what herding actually depends on, two parameter sweeps testing how sensitive the herding index is to herding-strength and vision-radius, and one final run driven entirely by real historical Bitcoin sentiment and price data, to check whether the patterns seen under synthetic conditions also hold up against actual market behavior. My role centered on the experimental side of this: setting up all six BehaviorSpace configurations correctly, running the full 570-run suite, and producing the structured outputs the rest of the group’s analysis and writing depend on.

D. T. K. S. Adhikari – s16664. Submodel Implemented or Analyzed. Conducted the quantitative time-series analysis of the agent-based model outputs, evaluating price movements, herding behavior, and market volatility across all six simulation scenarios (Baseline, SentimentShock, HerdingSensitivity, VisionRadiusSweep, RandomSentimentComparison, and Dataset).

Analysis or Visualization Contribution. Processed and merged the raw BehaviorSpace outputs exported from NetLogo using Python in Google Colab. Produced the descriptive summary tables, correlation matrices, line charts, box plots, and correlation heatmaps used throughout Section 5. Performed return-distribution analysis, including per-tick volatility calculations, outlier detection, and excess-kurtosis measurements computed both on the full tick series and after filtering out ticks affected by price-floor rebound artifacts ($\text{price} \leq 15$), in order to distinguish genuine fat-tailed market behavior from simulation boundary effects. Also computed the Pearson correlation between herding-strength and mean volatility within the Herding Sensitivity sweep, and subsequently added 95% confidence intervals and formal inferential tests (Welch’s t -tests, one-way ANOVA) on per-run averaged data to assess the statistical significance of cross-scenario differences reported in Table 3.

Writing Contribution. Wrote the quantitative findings underlying Subsections 5.1 to 5.3, 5.5, and 5.6 of the Results section, including the descriptive statistics, excess-kurtosis, and correlation analysis reported in Tables 3 to 5. These figures were subsequently reviewed together with A. J. M. M. P. Bandara to confirm that the Discussion section’s interpretation of the mechanisms driving volatility and fat tails was consistent with the underlying statistics.

Understanding of Project. The quantitative analysis carried out for this project showed clear differences in market behavior across the simulated scenarios. Under the Baseline scenario, price did not follow a steady linear path: it dipped to roughly 90 around tick 40 before rising sharply to approximately 162 by tick 150, giving a mean price of 117.256 (95% CI [82.28, 152.24], $n = 30$ runs), while the herding index remained comparatively stable, averaging 0.724. The SentimentShock scenario did not exhibit the same late-run price rise, instead remaining within a much narrower band for most of the simulation, producing a numerically lower mean price (94.589) and numerically higher mean volatility (0.026 vs. 0.022 for Baseline), together with a reduction in mean herding index to 0.664; however, neither the price nor the volatility difference between Baseline and SentimentShock reached statistical significance at the run level (Welch’s $t = 1.25$, $p = .221$ for price; $t = -1.08$, $p = .285$ for volatility), indicating that these two scenarios are not reliably separable on mean price or volatility alone despite their visibly different trajectories. A one-way ANOVA confirmed that mean volatility does differ significantly across the full set of four directly compared scenarios ($F = 7.64$, $p = .0001$), consistent with RandomSentimentComparison’s markedly tighter distribution. Return-distribution analysis identified pronounced fat-tailed behavior in the Baseline scenario, with an excess kurtosis of 451.827 in the unfiltered data that fell to 24.531 once ticks affected by price-floor rebounds were excluded, confirming that low-price boundary effects substantially exaggerated tail thickness while genuine extreme market movements still remained well above the levels observed in the control scenarios. Correlation analysis within the Herding Sensitivity sweep further identified a statistically significant, moderate positive relationship ($r = 0.345$, $p = 4.27 \times 10^{-8}$) between the herding-strength parameter and mean volatility, indicating that greater susceptibility to crowd behavior is associated with

increased price fluctuations, though the effect size is modest rather than large. This quantitative evidence directly underpins the group’s central finding that peer-driven crowd imitation — evident in the fat tails and volatility clustering statistics rather than in the raw mean-volatility level — is the dominant source of instability in the model, distinct from sentiment content alone.

A. J. M. M. P. Bandara – s16670. Submodel Implemented or Analyzed. Interpreted the results of the agent-based model by relating them to its underlying demand mechanism (Equation (4.1)) and placing the findings within the existing agent-based finance literature. This included explaining how the model’s main results—such as the strong fat tails in the Baseline scenario, the modest positive relationship between herding strength and volatility, and the near-random behaviour of the real sentiment dataset—could be understood through the model’s structure, particularly the separate roles of the sentiment and crowd-following terms in the chartists’ demand equation, rather than simply reporting the numerical results.

Analysis or Visualization Contribution. Conducted the literature review for the Comparison to Previous Work section (Subsection 6.3) by identifying and reviewing three relevant studies—Hessary and Hadzikadic [4], Lux and Marchesi [5], and Bokhari [2]. Compared the group’s findings with these studies to identify similarities and differences, and cross-checked all discussion-related claims, including the herding-strength correlation, kurtosis, and volatility autocorrelation results, against the corresponding tables and figures in Section 5 to ensure that the interpretation accurately reflected the reported results.

Writing Contribution. Wrote the entire Discussion and Conclusion section of the report (Section 6), including the Summary of Main Findings (Subsection 6.1), Interpretation and Implications (Subsection 6.2) with the Mechanisms Underlying Model Behavior and Connections to Real-World Phenomena subsections, Comparison to Previous Work (Subsection 6.3), Limitations (Subsection 6.4), Future Directions (Subsection 6.5), and the Conclusion (Subsection 6.6).

Understanding of Project. This project aimed to investigate whether social media sentiment drives investor herding and can reproduce two well-known features of financial markets: fat-tailed returns and volatility clustering. My main contribution was writing the Discussion section, where I explained the reasons behind the results rather than simply describing them. The model showed that the crowd-following behaviour of investors had a much stronger effect on market volatility than social media sentiment. Since the crowd-following mechanism was active in every scenario, including the Baseline scenario where there was no sentiment, investors continued to react to each other’s decisions, creating feedback loops that produced fat tails and volatility clustering. This was different from what our group originally expected. I also compared these findings with previous studies, particularly the work of Lux and Marchesi, who showed that market instability can emerge from investors imitating one another even without external information, and Hessary and Hadzikadic, whose results also linked herding with market volatility. Finally, I discussed the limitations of the model, including the price-floor artifact, the difficulty of separating the effects of sentiment and crowd-following, and the limited calibration using real Bitcoin data, to ensure that the conclusions were realistic and appropriately qualified.

REFERENCES

- [1] D. Araci, *FinBERT: Financial Sentiment Analysis with Pre-trained Language Models*, arXiv preprint arXiv:1908.10063, 2019.
- [2] A. Bokhari, *Unpacking Herding in Crypto Markets: An Agent-Based Study of True and Spurious Herding Dynamics*, 2025.
- [3] R. Cont, *Empirical properties of asset returns: stylized facts and statistical issues*, Quantitative Finance, 1 (2001), pp. 223–236.
- [4] Y. K. Hessary and M. Hadzikadic, *An agent-based study of herding relationships with financial markets phenomena*, in Proceedings of the 2017 Winter Simulation Conference (WSC), IEEE, 2017, pp. 1204–1215.
- [5] T. Lux and M. Marchesi, *Volatility clustering in financial markets: a micro-simulation of interacting agents*, International Journal of Theoretical and Applied Finance, 3 (2000), pp. 675–702.